meaning it will receive the full gradient update, while earlier moves will be discounted by a fraction such as 0.5 so the gradient steps will be smaller.

Let's add those into our objective (loss) function. The final objective function that we will tell PyTorch to minimize is $-\gamma_t * G_t * \log\pi_s(a|\theta)$. Remember, γ_t is the discount factor, and the subscript *t* tells us its value will depend on the time step *t*, since we want to discount more distant actions more than more recent ones. The parameter G_t is called the *total return*, or *future return*, at time step *t*. It is the return we expect to collect from time step *t* until the end of the episode, and it can be approximated by adding the rewards from some state in the episode until the end of the episode.

$$G_t = r_t + r_{t+1} \ldots + r_{T-1} + r_T$$

Actions temporally more distant from the received reward should be weighted less than actions closer. If we win a game of Gridworld, the sequence of discounted rewards from the start to the terminal state might look something like [0.970, 0.980, 0.99, 1.0]. The last action led to the winning state of +1, and it is not discounted at all. The previous action is assigned a scaled reward by multiplying the terminal reward with the γ_{t-1} discount factor, which we've set to 0.99.

The discount is exponentially decayed from 1, $\gamma_t = \gamma_0^{(T-t)}$, meaning that the discount at time *t* is calculated as the starting discount (here 0.99) exponentiated to the integer time distance from the reward. The length of the episode (the total number of time steps) is denoted *T*, and the local time step for a particular action is *t*. For $T - t = 0$, $\gamma_{T-0} = 0.99^0 = 1$. For $T - t = 2$, the $\gamma_{T-2} = 0.99^2 = 0.9801$, and so on. Each time step back, the discount factor is exponentiated to the distance from the terminal step, which results in an exponential decay of the discount factor the more distant (and irrelevant) the action was to the reward outcome.

For example, if the agent is in state S_0 (i.e., time step $t = 0$) and it takes action a_1 and receives reward $t_{t+1} = -1$, the target update will be $-\gamma^0(-$